

# Pellevoisin: the Raspberry Pi 4 header

Pin 1 top left with the board's edge at the top: odd pins in the inner row, even pins in the outer row next to the edge. Highlighted cells are this organ's connections. Every GND pin is the Pi's ground, and nothing on the organ's side ever touches this header.

| ours                                    | function            | pin       | pin       | function             | ours                                     |
|-----------------------------------------|---------------------|-----------|-----------|----------------------|------------------------------------------|
| <b>MIDI: 6N137 module IN+</b>           | <b>3.3 V</b>        | <b>1</b>  | <b>2</b>  | 5 V                  | free (Pi fed by USB-C from the SD-25A-5) |
| free; I2C stays off                     | GPIO 2 / SDA        | <b>3</b>  | <b>4</b>  | 5 V                  | free                                     |
| <b>ON/OFF SWITCH, one side</b>          | <b>GPIO 3 / SCL</b> | <b>5</b>  | <b>6</b>  | <b>GND</b>           | <b>ON/OFF SWITCH, other side</b>         |
|                                         | GPIO 4              | <b>7</b>  | <b>8</b>  | <b>GPIO 14 / TXD</b> | <b>MIDI: 6N137 module IN-</b>            |
| <b>RESET opto cathode (PC817 pin 2)</b> | <b>GND</b>          | <b>9</b>  | <b>10</b> | GPIO 15 / RXD        | free                                     |
| <b>RESET: 330 ohm to PC817 anode</b>    | <b>GPIO 17</b>      | <b>11</b> | <b>12</b> | GPIO 18              |                                          |
| <b>STATUS LED: 330 ohm to LED anode</b> | <b>GPIO 27</b>      | <b>13</b> | <b>14</b> | <b>GND</b>           | <b>STATUS LED cathode</b>                |
|                                         | GPIO 22             | <b>15</b> | <b>16</b> | GPIO 23              |                                          |
|                                         | 3.3 V               | <b>17</b> | <b>18</b> | <b>GPIO 24</b>       | <b>PUMP: SSR terminal 3 (+)</b>          |
|                                         | GPIO 10             | <b>19</b> | <b>20</b> | <b>GND</b>           | <b>PUMP: SSR terminal 4 (-)</b>          |
|                                         | GPIO 9              | <b>21</b> | <b>22</b> | GPIO 25              |                                          |
|                                         | GPIO 11             | <b>23</b> | <b>24</b> | GPIO 8               |                                          |
|                                         | GND                 | <b>25</b> | <b>26</b> | GPIO 7               |                                          |
| never                                   | <b>ID_SD</b>        | <b>27</b> | <b>28</b> | <b>ID_SC</b>         | never                                    |
|                                         | GPIO 5              | <b>29</b> | <b>30</b> | GND                  |                                          |
|                                         | GPIO 6              | <b>31</b> | <b>32</b> | GPIO 12              |                                          |
|                                         | GPIO 13             | <b>33</b> | <b>34</b> | GND                  |                                          |
|                                         | GPIO 19             | <b>35</b> | <b>36</b> | GPIO 16              |                                          |
|                                         | GPIO 26             | <b>37</b> | <b>38</b> | GPIO 20              |                                          |
|                                         | GND                 | <b>39</b> | <b>40</b> | GPIO 21              |                                          |

Each line is paired with its nearest ground so every cable is a short twisted pair: 5 and 6 for the switch, 11 and 9 for reset, 13 and 14 for the status LED, 18 and 20 for the pump. The MIDI pair is pins 1 and 8 and carries no ground: TX is its return. Pins 3 and 5 are the I2C bus, which stays disabled so GPIO 3 is free for the switch and wakes a halted Pi when pulled low. The DSI screen uses its own ribbon connector.

Part pinouts: PC817 pin 1 anode, 2 cathode, 3 emitter, 4 collector (dot at pin 1). SSR-xx DA: 1 and 2 load (mains), 3 positive and 4 negative input, 3 to 32 V DC. 6N137 module: IN+ and IN- marked on the terminals; VCC, OUT, GND on the other side. SD-25A-5: +Vin/-Vin from the 12 V bus through a 2 A fuse, +V/-V trimmed to 5.1 V at the Pi into its USB-C; 1500 V isolation inside. organ\_web: --pump 24 --power-switch 3; reset on 17. Status LED: config.txt dtoverlay=gpio-led,gpio=27,label=pellevoisin,trigger=heartbeat; it beats while Linux runs and is dark when halted.

# Pellevoisin: the box, wired

Two halves in one aluminium case, and exactly four things cross between them, three by light and one by a transformer: the MIDI opto, the reset opto, the SSR, and the isolated converter that makes the Pi's 5 V from the 12 V bus.

```
MAINS -- IEC inlet with fuse +- E ---- earth stud on the case ---- motor frame
+- N -----+----- motor N
|
|          ATX N
+- L -- PANEL SWITCH, 2 poles, 3 positions, centre off, motor rated
      pole A common +- BOOK -----> motor L
                    +- AUTO -> SSR 1 -- SSR 2 -----> motor L
      pole B common --- AUTO -> ATX L (PS_ON tied to ATX GND: on with Auto)
      (OFF: neither pole connected; the ATX fan cools the box in Auto)
```

|                                                      |                                                           |
|------------------------------------------------------|-----------------------------------------------------------|
| ORGAN SIDE (the ATX's ground)                        | PI SIDE (the converter's output ground)                   |
| ATX 12 V, several yellows -> 12 V bus bar            | SD-25A-5 +V / -V, trimmed to 5.1 V --USB-C--> Pi 4        |
| ATX GND, several blacks --> organ GND bus bar        | DSI ribbon --> touchscreen                                |
| ATX PS_ON (green) -----> ATX GND: on with Auto       | GPIO 3 (pin 5) -- antique switch -- Pi GND (pin 6) on/off |
| 12 V bus / GND bus -----> boards 1-4, 5 A fuses      | GPIO 14 TX (pin 8) -- 6N137 IN- MIDI                      |
| 12 V bus -[2 A fuse]-> SD-25A-5 +Vin, GND bus > -Vin | 3.3 V (pin 1) ----- 6N137 IN+                             |
| ----- transformer inside the SD-25A-5 -----          | GPIO 24 (pin 18) -- SSR 3 (+) pump                        |
| last board ISP pin 2 -----> RJ12 pin 6 = 5 V bus     | Pi GND (pin 20) --- SSR 4 (-)                             |
| RJ12 pin 1 = /RESET bus (jumped board to board)      | GPIO 27 (pin 13) -[330 ohm]- LED - Pi GND (pin 14) alive  |
| RJ12: MIDI signal, GND, as shipped                   | GPIO 17 (pin 11) -[330 ohm]- PC817 anode reset            |
| 6N137 module: VCC < RJ12 pin 6, GND < RJ12 GND,      | Pi GND (pin 9) ----- PC817 cathode                        |
| OUT > boards' MIDI in                                |                                                           |
| PC817: collector - RJ12 pin 1, emitter - RJ12 GND    |                                                           |
| SSR 1 / 2: on the mains side, above; MOV across them |                                                           |

## Rules

- No wire, bar or chassis lug between the organ's GND bus and the Pi's ground: the Pi's 5 V comes through the converter's transformer, never from a buck. The earth stud carries earth only, and no DC negative goes to it.
- Mains bus bars shrouded, or covered DIN terminal blocks instead. Low-voltage bars may be bare.
- Mains and low voltage on opposite sides of a partition: inlet, switch, supplies' inputs, SSR 1 and 2, motor lead on one side; Pi, RJ12 bus, optos, SSR 3 and 4 on the other. A wire that must cross does so once, at right angles.
- The SSR's heat sink in the ATX fan's stream, thermal paste under the module. A varistor across SSR 1 and 2.
- Only the live is switched, ever. Neutral and earth run straight through. The panel switch is motor rated: 1 HP at 125 V, or 20 A. The antique on/off switch carries microamps and may be anything.
- Book: the motor runs, nothing else has power. Auto: the ATX starts at once (PS\_ON tied to its ground) and the 12 V bus, the fan, the boards and the Pi through its converter come up together; the motor runs only when the SSR says so. Off: everything isolated (the SSR alone leaks a milliamp).
- Before the panel goes from Auto to Off: the antique switch off, or Shut down on the Service tab, and wait for the status LED to go dark.